

Kevin Huang

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ACADEMIC BACKGROUND

PhD Electrical Engineering September 2022 - Present
University of Southern California, Los Angeles, CA

M.S. Computer Science Sept 2020 - June 2021
University of California San Diego, La Jolla, CA
• BS/MS with emphasis on artificial intelligence.

B.S. Computer Science Sept 2016 - June 2020
University of California San Diego, La Jolla, CA
• Minors: Mathematics, Music Computing & Arts

EMPLOYMENT HISTORY

Research Assistant August 2023 – Present
University of Southern California, Los Angeles, CA
• RA in the Signal Analysis and Interpretation Laboratory at USC.
• Primarily working on research studying speech production in the interdisciplinary SPAN research group. Also involved in multiple projects developing speech analysis pipelines.

Teaching Assistant Sept 2020 – March 2021
University of California San Diego, La Jolla, CA
• TA under the computer science department at UCSD for two quarters
• Responsibilities include preparing for discussion sections, holding office hours, and grading

Tutor April 2018 – March 2020
University of California San Diego, La Jolla, CA
• Tutor under the computer science department at UCSD for seven quarters and two summer sessions
• Responsibilities include holding lab/office hours, grading homeworks and tests, and helping communicate course content to students

Intern June 2018 - September 2018
Webroot Cybersecurity, San Diego, CA
• Summer internship working with a small team of 4 peers to revamp a DNSP agent distribution portal from scratch
• Used an Angular front-end with a Node.js back-end connected to a MySQL database

Student Worker Summer 2017
San Joaquin County Office of Education, Stockton, CA
• Helped teach Lego robotics summer camps
• Volunteered throughout the five prior years, which in addition to helping teach after-school and summer classes, helped perform various maintenance tasks around the robotics lab

TEACHING EXPERIENCE

TA - UCSD

- Fall 2020: CSE 167 - Computer Graphics
- Winter 2021: CSE 169 - Computer Animation

Tutor - UCSD

- Spring 2018: CSE 140L - Digital Systems Lab
- Fall 2018: CSE 20 - Discrete Math
- Fall 2018: CSE 167 - Computer Graphics
- Winter 2019: CSE 140(L) - Digital Systems
- Spring 2019: CSE 140L - Digital Systems Lab
- Summer 2019 I & II: CSE 140 - Digital Systems
- Fall 2019: CSE 167 - Computer Graphics
- Winter 2020: CSE 167 - Computer Graphics

PAPERS

Publications

Accepted

- Lee, J. *, Razmara, P. *, **Huang K. ***, et al. “”An Approach to Simultaneous Acquisition of Real-Time MRI Video, EEG, and Surface EMG for Articulatory, Brain, and Muscle Activity During Speech Production.” Accepted to Interspeech 2026.

Published

- Foley, S., Lee, J., **Huang, K.**, Shi, X., Lee, Y., Goldstein, L., & Narayanan, S. “”A long-form single-speaker real-time MRI speech dataset and benchmark.” ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2026.
- Lee, J., Foley, S., Lertpetchpun T., **Huang, K.**, Lee, Y. , Feng, T., Goldstein, L. , Byrd, D. & Narayanan, S. ”Arti-6: Towards six-dimensional Articulatory Speech Encoding.” ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2026.
- Feng, T., **Huang, K.**, Xu, A., Shi, X., Lartpetchpun, T., Lee, J., Lee, Y., Byrd, D., Narayanan, S. “Voxlect: A Speech Foundation Model Benchmark for Modeling Dialects and Regional Languages Around the Globe”, Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 1. 2026.
- **Huang, K.** , Foley, S. , Lee, J. , Lee, Y. , Byrd, D. & Narayanan, S. “On the Relationship between Accent Strength and Articulatory Features”, Submitted to Interspeech 2025. **Best Student Paper Award**

- Lee, J. , **Huang, K.** , Avramidis, K. , Pistrosch, S. , Gonzalez-Machorro, M. , Lee, Y. , Schuller, B. , Goldstein, L. & Narayanan, S. “Articulatory Feature Prediction from Surface EMG during Speech Production”, Submitted to Interspeech 2025.
- Zhang, Y. , Kumar, K. , Tian, Y. , Zhao, Z. , Shi, X. , **Huang, K.** , Lee, K. , Hsu, H. , Narayanan, S. , Nayak, K. & Goldstein, L. “Co-registration of Real-time MRI and Respiration for Speech Research ”, Submitted to Interspeech 2025.
- Kadiri, S. , **Huang, K.** , Hagedorn, C. , Byrd, D. , Alku, P. & Narayanan, S. “Formant Tracking by Combining Deep Neural Network and Linear Prediction”, IEEE Open Journal of Signal Processing
- Nguyen, H. , Foley, S. , **Huang, K.** , Shi, X. , Feng, T. & Narayanan, S. “Speech2rtMRI: Speech-Guided Diffusion Model for Real-time MRI Video of the Vocal Tract during Speech”, ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)
- Shi, X. , Feng, T. , **Huang, K.** , Kadiri, S. , Lee, J. , Lu, Y. , Zhang, Y. , Goldstein, L. & Narayanan, S. “Interpreting the Performance of Pre-trained Speech Models on Diverse Pronunciation Styles using Direct Articulatory Information”, JASA Express Letters 4.11 (2024).
- **Huang, K.** , Goldberg, J. , Goldstein, L. & Narayanan, S. “Analysis of articulatory setting for L1 and L2 English speakers using MRI data”, Interspeech 2024. **Best Student Paper Award**
- Xu, A. , **Huang, K.** , Feng T. , Lue, S. , Tager-Flusberg, H. & Narayanan, S. “Exploring Speech Foundation Models for Speaker Diarization in Child-Adult Dyadic Interactions”, Interspeech 2024
- Xu, A. , **Huang, K.** , Feng T. , Tager-Flusberg, H. & Narayanan, S. “Audio-visual child-adult speaker classification in dyadic interactions”, ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)
- Dubnov, S. , Chen, K. & **Huang, K.** “Deep Music Information Dynamics”, Journal of Creative Music Systems, 2022

Course Related

- “Exploring brain and articulatory dynamics during speech production” for PSYC 599 - Computational Language Neuroscience
- “A brief look into the effects of Nonuniform Time-Scale Modification of Speech Signals on Transformer-based Automatic Speech Recognition towards the study of Prosody” for LING 487 - Speech Synthesis and Recognition
- “Text2RIR” for EE 522 - Immersive Audio Signal Processing
- “Towards Cross-Cultural Analysis using Music Information Dynamics,” for CSE 298 - Independent Study, 2021
- “Sparsity in Nonnegative Matrix Factorization,” for CSE 205 - Convex Optimization Algorithms, 2021.
- “Mel2Mel: A C-GAN architecture to improve the quality of parametric singing voice synthesis,” for MUS 206 - Experimental Studies Seminar, 2020.
- “Evaluating Frequency-Based Fake Object Detection Models in Practice” for CSE 253 - Neural Networks/Pattern Recognition, 2020.

SKILLS

- Programming Languages: Python, C++, Javascript, C#, C, Java, Rust
- Libraries: PyTorch, OpenGL, Tensorflow
- Software: Unity, Blender, Pure Data, Ableton